

Proofs and Modern Mathematics Theorems List

Logic

- For statements P and Q , $P \Rightarrow Q$ and $(\sim P) \vee Q$ are logically equivalent.
- For statements P , Q , and R , $P \vee (Q \wedge R)$ and $(P \vee Q) \wedge (P \vee R)$ are logically equivalent.
- For statements P and Q , $\sim (P \vee Q)$ and $(\sim P) \wedge (\sim Q)$ are logically equivalent.
- (Modus Ponens) For statements P and Q , $(P \wedge (P \Rightarrow Q)) \Rightarrow Q$ is a tautology.
- (Modus Tollens) For statements P and Q , $(P \Rightarrow Q) \wedge (\sim Q) \Rightarrow (\sim P)$ is a tautology.
- (Syllogism) For statements P , Q , and R , $((P \Rightarrow Q) \wedge (Q \Rightarrow R)) \Rightarrow (P \Rightarrow R)$ is a tautology.

Modular Arithmetic

- For odd integers a and b and any integer c , $ac + bc$ is even.
- For any integer n , $n^2 + n$ is even.
- For odd integers a and b , $a^2 + b^2$ is even.
- An integer n is even if and only if $(n + 1)^2$ is odd.
- For an integer n , $3n - 1$ is even if and only if $5n - 2$ is odd.
- Every odd integer is the difference of two squares.
- Every square number is the sum of consecutive odd numbers.
- A sum of two integers is even if and only if the numbers have the same parity.
- A product of two integers is odd if and only if the two numbers are odd.
- If the product and the sum of two integers are both even, then both integers are even.
- For integers a and b , if ab is odd, then $a^4 + b^4$ is even.
- The number 200 cannot be expressed as a sum of an odd integer and two even integers.
- For integers a , b , c , and n , if $a \equiv b \pmod{n}$ and $a \equiv c \pmod{n}$, then $b \equiv c \pmod{n}$.
- For integers a , b , c and n , if $a \equiv b \pmod{n}$, then $a + c \equiv b + c \pmod{n}$ and $ca \equiv cb \pmod{n}$.
- For integers a , b , and c , if a divides b and b divides c , then a divides c .
- For integers a , b , and c , if $ac|bc$ and $c \neq 0$, then $a|b$.
- For an integer a and distinct prime p and q , If $p|a$ and $q|a$, then $pq|a$.
- For an integer a and a prime p , if $a \equiv 1 \pmod{p}$, $a^2 \equiv 1 \pmod{p}$.
- For integers a , b , and n , if $a \equiv b \pmod{n}$, then $a^2 \equiv b^2 \pmod{n}$.
- (The Division Algorithm) For integers a and $b > 0$, there exist unique integers q and r such that $a = bq + r$ where $0 \leq r < b$.

- For an integer a and a prime p , if $p \nmid a$, then the greatest common divisor of a and p is 1.
- For integers a , b , and c and all integers m and n , if $a \mid b$ and $a \mid c$, then $a \mid (bm + cn)$.
- (Bézout's Identity) For positive integers a and b , the greatest common divisor of a and b can be expressed as $ax + by$ for some integers x and y .
- (Euclid's Lemma) For integers a and b and a prime p , if $p \mid ab$, then $p \mid a$ or $p \mid b$.
- For a prime p and an integer n , p divides n if and only if p divides n^2 .
- (Fermat's Little Theorem) For an integer a and a prime p , $a^p \equiv a \pmod{p}$ or $a^p - a$ is divisible by p .
- (Binomial Theorem) For integers a and b and a prime p , $(a + b)^p \equiv a^p + b^p \pmod{p}$.

Irrational Numbers

- (Pythagoras) The number $\sqrt{2}$ is irrational.
- The square root of a prime number is irrational.
- (Euclid) There are an infinite number of prime numbers.
- The sum of a rational number and an irrational number is irrational.
- There exist irrational numbers a and b such that ab is rational.
- There exist irrational numbers a and b such that a^b is rational.
- The product of an irrational number and a non-zero rational number is irrational.
- The number $\sqrt{6}$ is irrational.
- The sum $\sqrt{2} + \sqrt{3}$ is irrational.
- (The Archimedean Property) For any positive real numbers ε , there exist natural numbers n, k such that $\frac{1}{n} < \varepsilon$ and $\varepsilon < k$.
- For any real number c , there a unique integer k in the interval $[c, c + 1)$.
- The rational numbers are dense in the reals.
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- A set S is dense in $\mathbb{R}$ if and only if every real number x is a limit of a sequence in S .

Mathematical Induction

- For a natural n , $\sum_{i=1}^n i = \frac{n(n+1)}{2}$.
- For a natural n , $\sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6}$.
- For a natural n , $\sum_{i=1}^n 2^i = 2^{n+1} - 1$.

- For a non-negative integer n , the n^{th} derivative of e^{x^2} is $p(x)e^{x^2}$ where $p(x)$ is an n^{th} degree polynomial.
- For a non-negative integer n , $n! = \int_0^\infty x^n e^{-x} dx$.
- Every positive integer is a unique product of prime numbers.
- Every product of a finite collection of odd numbers is odd.
- Every polynomial is a unique product of irreducible polynomials.

Set Operations

- For sets A , B , and C , if $A \subset B$ and $B \subset C$, then $A \subset C$.
- For sets A and B , $A \cup B = A \cap B$ if and only if $A = B$.
- For sets A , B , and C , if $A \cup B = A \cup C$ and $A \cap B = A \cap C$, then $B = C$.
- For sets A , B , and C , $(A \setminus B) \cup (A \cap C) = A \setminus (B \cap C)$.
- For sets A , B , and C , $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$.
- For sets A and B , $A^C \setminus B^C = B \setminus A$.
- (De Morgan's Laws) For sets A and B , $(A \cup B)^C = A^C \cap B^C$ and $(A \cap B)^C = A^C \cup B^C$.
- For sets A , B , and C , $(A \triangle B) \cap C = (A \cap C) \triangle (B \cap C)$.
- For sets A , B , and C , $(A \cup B) \triangle C = (A \triangle C) \triangle (B \setminus A)$.

Product, Power, and Indexed Sets

- For sets A , B , and C , $A \times (B \cap C) = (A \times B) \cap (A \times C)$.
- For a set A , $|P(A)| = 2^{|A|}$.
- For sets A and B , $A \subset B$ if and only if $P(A) \subset P(B)$.
- For sets A and B , $\mathcal{P}(A) \cup \mathcal{P}(B) \subset \mathcal{P}(A \cup B)$.
- For sets A and B , $\mathcal{P}(A) \cap \mathcal{P}(B) = \mathcal{P}(A \cap B)$.
- For sets A , B , C , and D , $(A \times B) \cup (C \times D) \subset (A \cup C) \times (B \cup D)$.
- For a countable family of sets $\{A_i\}_i$, $\bigcap_{i \in I} A_i^C = \left(\bigcup_{i \in I} A_i \right)^C$.
- For a countable family of sets $\{A_i\}_i$ and a set S , $\bigcup_{i \in I} A_i \times S = \bigcup_{i \in I} (A_i \times S)$.

Measurable Sets

- (Monotonicity) For measurable sets E and F , if $E \subset F$, then $\mu(E) \leq \mu(F)$.
- (Subadditivity) For a family of measurable sets $\{E_i\}_i$, $\mu(\bigcup_1^\infty E_i) \leq \sum_1^\infty \mu(E_i)$.

- (Continuity from Below) For a family of measurable sets $\{E_i\}_i$ such that $E_1 \subset E_2 \subset \dots$, $\mu(\cup_1^\infty E_i) = \lim_{i \rightarrow \infty} \mu(E_i)$.
- (Continuity from Above) For a family of measurable sets $\{E_i\}_i$ such that $E_1 \supset E_2 \supset \dots$ where $\mu(E_1) < \infty$, $\mu(\cap_1^\infty E_i) = \lim_{i \rightarrow \infty} \mu(E_i)$.
- If E and F are measurable sets, $\mu(E) + \mu(F) = \mu(E \cup F) + \mu(E \cap F)$.
- If E is a measurable set, $\mu_E(A) = \mu(A \cap E)$ is a measure.
- If $\{\mu\}_i$ is a family of measures and $\{a_i\}_i$ is a family of non-negative real numbers, $\sum_i^\infty a_i \mu_i$ is a measure provided that the sum converges.

Functions

- For a function $f : X \rightarrow Y$ and subsets $A, B \subset X$, $f(A \cup B) = f(A) \cup f(B)$.
- For a function $f : X \rightarrow Y$ and subsets $A, B \subset X$, $f(A \cap B) \subset f(A) \cap f(B)$.
- For a function $f : X \rightarrow Y$ and subsets $A, B \subset X$, $f(A) \setminus f(B) \subset f(A \setminus B)$.
- For a function $f : X \rightarrow Y$ and subsets $A \subset X$ and $S \subset Y$, $f(A) \subset S$ if and only if $A \subset f^{-1}(S)$.
- For a function $f : X \rightarrow Y$ and subsets $S, T \subset Y$, $f^{-1}(S \cup T) = f^{-1}(S) \cup f^{-1}(T)$.
- For a function $f : X \rightarrow Y$ and subsets $S, T \subset Y$, $f^{-1}(S \cap T) = f^{-1}(S) \cap f^{-1}(T)$.
- For a function $f : X \rightarrow Y$ and subsets $S, T \subset Y$, $f^{-1}(S \setminus T) = f^{-1}(S) \setminus f^{-1}(T)$.
- For a function $f : X \rightarrow Y$, $X \subset f^{-1}(f(X))$ and $f(f^{-1}(Y)) = Y$.
- The inverse of a function is unique.
- For sets A and B , $A = B$ if and only if $\chi_A(x) = \chi_B(x)$.
- For sets A and B , $\chi_{A \cap B}(x) = \chi_A(x) \cdot \chi_B(x)$.
- For functions f and g , if $g \circ f$ is injective, f is injective.
- For functions f and g , if $g \circ f$ is surjective, g is surjective.
- The composition of bijective functions is bijective.
- For bijections $f : A \rightarrow B$ and $g : B \rightarrow A$, $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$.
- A function is bijective if and only if it has an inverse.
- For sets A and B both of cardinality n , there are $n!$ bijections between them.
- For every permutation σ there exists a smallest natural number n such that $\sigma^n = i_\sigma$.
- Every representation of a permutation as a composition of transpositions has the same parity of the number of transpositions.

Relations, Posets, and Binary Operations

- If an equality relation is defined on a set, the equivalence classes partition the set.
- Congruence modulo n is an equivalence relation on the integers.
- The subset relation on a power set is a partial ordering but not a total ordering.
- The divisibility relation on the positive integers is a partial ordering but not a total ordering.
- For a function $f : A \rightarrow B$, $a \sim b$ if and only if $f(a) = f(b)$ is an equivalence relation on A .
- The set of natural powers of a prime p is totally ordered by divisibility.
- The union operation on a power set is closed, commutative, associative, and has an identity.
- The intersection operation on a power set is closed, commutative, and associative, and has an identity.
- The symmetric difference operation on a power set is closed, commutative, and associative, has an identity, and has inverses.
- The set of permutations on a finite set under composition has inverses.
- Identity and inverse elements of a binary operation are unique.

Infinite Sets and Cardinality

- For a function $f : A \rightarrow B$, if f is injective, $|B| \geq |A|$, and if f is surjective, $|A| \geq |B|$.
- (The Schröder-Bernstein Theorem) If A and B are sets such that $|A| \leq |B|$ and $|B| \leq |A|$, then $|A| = |B|$.
- Every infinite subset of a countably infinite set is countably infinite.
- The finite product of countably infinite sets is countably infinite.
- The countable union of countably infinite sets is countably infinite.
- The rationals and the algebraic closure of the rationals are countably infinite.
- (Cantor's Theorem) The cardinality of the power set of a set is strictly greater than the cardinality of the set itself.
- The real interval $[0, 1]$ is uncountably infinite.
- The cardinality of the power set of the natural numbers is equal to the cardinality of the real numbers.
- (Zorn's Lemma) If every chain in a partially ordered set P has an upper bound in P , then P has a maximal element.
- (Well Ordering Principal) Every set can be well-ordered.
- (Axiom of Choice) For any collection of non-empty sets C , there exists a choice function f on C such that for every $A \in C$, $f(A) \in A$.